It has a remarkable property of setting with water. On mixing with an adequate quantity of water it forms a plastic mass that gets into a hard solid in 5 to 15 minutes.

**Uses:**

Plaster of Paris is used as/in,

1. The building industry as well as plasters.
2. For immobilising the affected part of organ where there is a bone fracture or sprain.
3. Employed in dentistry, in ornamental work and for making casts of statues and busts.



## 5.7 Biological importance of magnesium and calcium

Magnesium and calcium also plays a vital role in biological functions. A typical adult human body contains about 25 g of magnesium and 1200 g of calcium. Magnesium plays an important role in many biochemical reactions catalysed by enzymes. It is the co-factor of all enzymes that utilize ATP in phosphate transfer and energy release. It also essential for DNA synthesis and is responsible for the stability and proper functioning of DNA. It is also used for balancing electrolytes in our body. Deficiency of magnesium results into convulsion and neuromuscular irritation.

Calcium is a major component of bones and teeth. It is also present in in blood and its concentration is maintained by hormones (calcitonin and parathyroid hormone). Deficiency of calcium in blood causes it to take longer time to clot. It is also important for muscle contraction.

The main pigment that is responsible for photosynthesis, chlorophyll, contains magnesium which plays an important role in photosynthesis.



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